METRIC

MIL-DTL-5624X
12 February 2024
SUPERSEDING
MIL-DTL-5624W
28 March 2016

# DETAIL SPECIFICATION

# TURBINE FUEL, AVIATION, GRADE JP-5 (NATO F-44)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers only high flash point aviation turbine fuel NATO F-44 (JP-5) (see 6.1). Requirements for NATO F-40 (JP-4) have been removed from this specification without replacement. Synthesized hydrocarbons from non-conventional sources require specific guidance that is outside the scope of MIL-DTL-5624. This guidance is found in the latest revisions of ASTM D1655 and ASTM D7566.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, Mail Stop B120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to lke-navairdoccom.fct@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

| | |
|---|---|
| MIL-PRF-25017 | Inhibitor, Corrosion/Lubricity Improver, Fuel Soluble (NATO S-1747) (Metric) |
| MIL-DTL-85470 | Inhibitor, Icing, Fuel System, High Flash, NATO Code Number S-1745 (Metric) |

DEPARTMENT OF DEFENSE STANDARD

| | |
|---|---|
| MIL-STD-290 | Packaging and Marking of Petroleum and Related Products |

QUALIFIED PRODUCTS LIST

| | |
|---|---|
| QPL-25017 | Inhibitor, Corrosion/Lubricity Improver, Fuel Soluble (NATO S-1747) (Metric) |

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

| | |
|---|---|
| ASTM D56 | Standard Test Method for Flash Point by Tag Closed Cup Tester |
| ASTM D86 | Standard Test Method for Distillation of Petroleum Products and Liquid Fuels at Atmospheric Pressure |
| ASTM D93 | Standard Test Methods for Flash Point by Pensky-Martens Closed Cup Tester |
| ASTM D130 | Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test |
| ASTM D156 | Standard Test Method for Saybolt Color of Petroleum Products (Saybolt Chromometer Method) |
| ASTM D381 | Standard Test Method for Gum Content in Fuels by Jet Evaporation |
| ASTM D445 | Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity) |
| ASTM D976 | Standard Test Method for Calculated Cetane Index of Distillate Fuels |

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| ASTM D1298 | Standard Test Method for Density, Relative Density, or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method |
| ASTM D1319 | Standard Test Method for Hydrocarbon Types in Liquid Petroleum Products by Fluorescent Indicator Adsorption |
| ASTM D1322 | Standard Test Method for Smoke Point of Kerosene and Aviation Turbine Fuel |
| ASTM D1840 | Standard Test Method for Naphthalene Hydrocarbons in Aviation Turbine Fuels by Ultraviolet Spectrophotometry |
| ASTM D2276 | Standard Test Method for Particulate Contaminant in Aviation Fuel by Line Sampling |
| ASTM D2386 | Standard Test Method for Freezing Point of Aviation Fuels |
| ASTM D2622 | Standard Test Method for Sulfur in Petroleum Products by Wavelength Dispersive X-Ray Fluorescence Spectrometry |
| ASTM D2624 | Standard Test Methods for Electrical Conductivity of Aviation and Distillate Fuels |
| ASTM D2887 | Standard Test Method for Boiling Range Distribution of Petroleum Fractions by Gas Chromatography |
| ASTM D3227 | Standard Test Method for (Thiol Mercaptan) Sulfur in Gasoline, Kerosine, Aviation Turbine, and Distillate Fuels (Potentiometric Method) |
| ASTM D3241 | Standard Test Method for Thermal Oxidation Stability of Aviation Turbine Fuels |
| ASTM D3242 | Standard Test Method for Acidity in Aviation Turbine Fuel |
| ASTM D3338/ D3338M | Standard Test Method for Estimation of Net Heat of Combustion of Aviation Fuels |
| ASTM D3701 | Standard Test Method for Hydrogen Content of Aviation Turbine Fuels by Low Resolution Nuclear Magnetic Resonance Spectrometry |
| ASTM D3948 | Standard Test Method for Determining Water Separation Characteristics of Aviation Turbine Fuels by Portable Separometer |
| ASTM D4052 | Standard Test Method for Density, Relative Density, and API Gravity of Liquids by Digital Density Meter |
| ASTM D4057 | Standard Practice for Manual Sampling of Petroleum and Petroleum Products |
| ASTM D4177 | Standard Practice for Automatic Sampling of Petroleum and Petroleum Products |
| ASTM D4294 | Standard Test Method for Sulfur in Petroleum and Petroleum Products by Energy Dispersive X-Ray Fluorescence Spectrometry |

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| ASTM D4306 | Standard Practice for Aviation Fuel Sample Containers for Tests Affected by Trace Contamination |
| ASTM D4529 | Standard Test Method for Estimation of Net Heat of Combustion of Aviation Fuels |
| ASTM D4737 | Standard Test Method for Calculated Cetane Index by Four Variable Equation |
| ASTM D4809 | Standard Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter (Precision Method) |
| ASTM D4952 | Standard Test Method for Qualitative Analysis for Active Sulfur Species in Fuels and Solvents (Doctor Test) |
| ASTM D5006 | Standard Test Method for Measurement of Fuel System Icing Inhibitors (Ether Type) in Aviation Fuels |
| ASTM D5291 | Standard Test Methods for Instrumental Determination of Carbon, Hydrogen, and Nitrogen in Petroleum Products and Lubricants |
| ASTM D5452 | Standard Test Method for Particulate Contamination in Aviation Fuels by Laboratory Filtration |
| ASTM D5453 | Standard Test Method for Determination of Total Sulfur in Light Hydrocarbons, Spark Ignition Engine Fuel, Diesel Engine Fuel, and Engine Oil by Ultraviolet Fluorescence |
| ASTM D5972 | Standard Test Method for Freezing Point of Aviation Fuels (Automatic Phase Transition Method) |
| ASTM D6045 | Standard Test Method for Color of Petroleum Products by the Automatic Tristimulus Method |
| ASTM D6379 | Standard Test Method for Determination of Aromatic Hydrocarbon Types in Aviation Fuels and Petroleum Distillates—High Performance Liquid Chromatography Method with Refractive Index Detection |
| ASTM D6890 | Standard Test Method for Determination of Ignition Delay and Derived Cetane Number (DCN) of Diesel Fuel Oils by Combustion in a Constant Volume Chamber |
| ASTM D6986 | Standard Test Method for Free Water, Particulate and Other Contamination in Aviation Fuels (Visual Inspection Procedures) |
| ASTM D7042 | Standard Test Method for Dynamic Viscosity and Density of Liquids by Stabinger Viscometer (and the Calculation of Kinematic Viscosity) |
| ASTM D7153 | Standard Test Method for Freezing Point of Aviation Fuels (Automatic Laser Method) |
| ASTM D7154 | Standard Test Method for Freezing Point of Aviation Fuels (Automatic Fiber Optical Method) |
| ASTM D7171 | Standard Test Method for Hydrogen Content of Middle Distillate Petroleum Products by Low-Resolution Pulsed Nuclear Magnetic Resonance Spectroscopy |

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| ASTM D7524 | Standard Test Method for Determination of Static Dissipater Additives (SDA) in Aviation Turbine Fuel and Middle Distillate Fuels—High Performance Liquid Chromatography (HPLC) Method |
| ASTM D7566 | Standard Specification for Aviation Turbine Fuel Containing Synthesized Hydrocarbons |
| ASTM D7619 | Standard Test Method for Sizing and Counting Particles in Light and Middle Distillate Fuels, by Automatic Particle Counter |
| ASTM D7668 | Standard Test Method for Determination of Derived Cetane Number (DCN) of Diesel Fuel Oils—Ignition Delay and Combustion Delay Using a Constant Volume Combustion Chamber Method |
| ASTM D7797 | Standard Test Method for Determination of the Fatty Acid Methyl Esters Content of Aviation Turbine Fuel Using Flow Analysis by Fourier Transform Infrared Spectroscopy-Rapid Screening Method |
| ASTM D7945 | Standard Test Method for Determination of Dynamic Viscosity and Derived Kinematic Viscosity of Liquids by Constant Pressure Viscometer |
| ASTM D8267 | Standard Test Method for Determination of Total Aromatic, Monoaromatic and Diaromatic Content of Aviation Turbine Fuels Using Gas Chromatography with Vacuum Ultraviolet Absorption Spectroscopy Detection (GC-VUV) |
| ASTM D8305 | Standard Test Method for the Determination of Total Aromatic Hydrocarbons and Total Polynuclear Aromatic Hydrocarbons in Aviation Turbine Fuels and other Kerosene Range Fuels by Supercritical Fluid Chromatography |
| ASTM E29 | Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications |

(Copies of these documents are available from https://www.astm.org/.)

ENERGY INSTITUTE

| | |
|---|---|
| IP 540 | Determination of the Existent Gum Content of Aviation Turbine Fuel - Jet Evaporation Method |
| IP 565 | Determination of the Level of Cleanliness of Aviation Turbine Fuel -- Portable Automatic Particle Counter Method |
| IP 577 | Determination of the Level of Cleanliness of Aviation Turbine Fuel-- Automatic Particle Counter Method Using Light Extinction |
| IP 583 | Determination of the Fatty Acid Methyl Esters Content of Aviation Turbine Fuel Using Flow Analysis by Fourier Transform Infrared Spectroscopy - Rapid Screening Method |
| IP 585 | Determination of Fatty Acid Methyl Esters (FAME), Derived from Bio-Diesel Fuel, in Aviation Turbine Fuel - GC-MS with Selective Ion Monitoring/Scan Detection Method |

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| IP 590 | Determination of Fatty Acid Methyl Esters (FAME) in Aviation Turbine Fuel - HPLC Evaporative Light Scattering Detector Method |
| IP 599 | Determination of Fatty Acid Methyl Esters (FAME) in Aviation Turbine Fuel - Gas Chromatography Using Heart-Cut and Refocusing |

(Copies of these documents are available from https://publishing.energyinst.org/.)

INTERNATIONAL STANDARDIZATION ORGANIZATION (ISO)

| | |
|---|---|
| ISO 4406 | Hydraulic fluid power – Fluids - Method for Coding the Level of Contamination by Solid Particles |
| ISO 11171 | Hydraulic fluid power – Calibration of Automatic Particle Counters for Liquids |

(Copies of these documents are available from https://www.iso.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Materials. Aviation turbine fuel is a complex mixture of hydrocarbons that varies depending on crude source and manufacturing process. JP-5 supplied under this specification shall be refined hydrocarbon distillate fuel oils, which contain additives in accordance with 3.3. Unless the feedstock from which the fuel is refined is as specified in 3.1.1 to 3.1.7, the feedstock shall be crude oils, natural gas liquid condensates, heavy oil, petroleum, oil sands, oil shale, or mixtures thereof.

3.1.1 Co-hydroprocessed synthesized kerosene. Hydrocarbons in the kerosene boiling range derived from non-petroleum feedstocks that have been processed simultaneously (co-processing) with hydrocarbons from conventional sources are permitted in the manufacture of JP-5. The co-hydroprocessed synthesized kerosene shall be produced from the feedstocks specified in 3.1.1.1 and 3.1.1.2 and shall not exceed the concentration stated in these sections. The finished fuel shall conform to the properties listed in tables I, II, III, and V and contain additives in accordance with 3.3 through 3.3.6.

3.1.1.1 Mono-, di-, and triglycerides, free fatty acids, and fatty acid esters. Co-processing of mono-, di-, and triglycerides, free fatty acids, and fatty acid esters with conventionally sourced hydrocarbons is acceptable for the manufacture of JP-5. The process streams used for jet fuel production in co-processing refinery units shall not exceed 5 vol% mono-, di-, and triglycerides, free fatty acids, and fatty acid esters in feedstock volume with the balance (≥95 vol%) being conventionally sourced hydrocarbons. Co-processing shall include hydrocracking or hydrotreating and fractionation. Processing may also include other conventional refinery processes. The finished fuel is limited to 5 vol% co-hydroprocessed

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

synthesized kerosene derived from mono-, di-, and triglycerides, free fatty acids, and fatty acid esters.

3.1.1.2 Hydrocarbons derived from synthesis gas via the Fischer-Tropsch (FT) process. Co-processing of hydrocarbons derived from synthesis gas via the Fischer-Tropsch (FT) process using iron or cobalt catalyst with conventionally sourced hydrocarbons is acceptable for the manufacture of JP-5. The process streams used for jet fuel production in co-processing refinery units shall not exceed 5 vol% FT hydrocarbons in feedstock volume with the balance (≥95 vol%) being conventionally sourced hydrocarbons. Co-processing shall include hydrocracking and fractionation. Processing may also include other conventional refinery processes. The finished fuel is limited to 5 vol% co-hydroprocessed synthesized kerosene derived from FT feedstock.

3.1.2 Fischer-Tropsch hydroprocessed synthesized paraffinic kerosene (FT-SPK). A maximum of 50 percent by volume of the finished fuel may consist of FT-SPK synthetic blending components. FT-SPK shall be comprised of hydroprocessed synthesized paraffinic kerosene wholly derived from paraffins and olefins produced from synthesis gas via the FT process using iron or cobalt catalyst. The FT-SPK synthetic blending components shall conform to the requirements in ASTM D7566 Annex A1. The finished fuel shall conform to the properties listed in tables I, II, IV and V and contain additives in accordance with 3.3 through 3.3.6.

3.1.3 Synthesized paraffinic kerosene from hydroprocessed esters and fatty acids (HEFA-SPK). A maximum of 50 percent by volume of the finished fuel may consist of HEFA-SPK synthetic blending components. HEFA-SPK shall be comprised of hydroprocessed synthesized paraffinic kerosene wholly derived from hydrogenation and deoxygenation of fatty acid esters and free fatty acids. The HEFA-SPK synthetic blending components shall conform to the requirements in ASTM D7566 Annex A2. The finished fuel shall conform to the properties listed in tables I, II, IV and V and contain additives in accordance with 3.3 through 3.3.6.

3.1.4 Synthesized iso-paraffins from hydroprocessed fermented sugars (SIP). A maximum of 10 percent by volume of the finished fuel may consist of SIP synthetic blending components derived from hydroprocessed fermented sugars. SIP blend components shall be comprised of hydroprocessed synthesized iso-paraffins wholly derived from farnesene produced from fermentable sugars. The SIP synthetic blending components shall conform to the requirements in ASTM D7566 Annex A3. The finished fuel shall conform to the properties listed in tables I, II, IV and V and contain additives in accordance with 3.3 through 3.3.6.

3.1.5 FT synthesized paraffinic kerosene plus aromatics. A maximum of 50 percent by volume of the finished fuel may consist of FT synthesized paraffinic kerosene plus aromatics (SPK/A) synthetic blending components. SPK/A blend components shall be comprised of FT-SPK as defined in 3.1.2 combined with synthesized aromatics from the alkylation of non-petroleum derived light aromatics (primarily benzene). The SPK/A synthetic blending components shall conform to the requirements in ASTM D7566 Annex A4. The finished fuel

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

shall conform to the properties listed in tables I, II, IV and V and contain additives in accordance with 3.3 through 3.3.6.

3.1.6 Alcohol-to-jet synthetic paraffinic kerosene (ATJ-SPK). A maximum of 50 percent by volume of the finished fuel may consist of ATJ-SPK synthetic blending components derived from ethanol; all other alcohol feedstocks such as isobutanol and isobutene are not approved. The ATJ-SPK synthetic blending components shall conform to the requirements in ASTM D7566 Annex A5. The finished fuel shall conform to the properties listed in tables I, II, IV, and V and contain additives in accordance with 3.3 through 3.3.6.

3.1.7 Synthesized paraffinic kerosene from hydroprocessed hydrocarbons, esters, and fatty acids (HC-HEFA). A maximum of 10 percent by volume of the finished fuel may consist of HC-HEFA synthetic blending components derived from hydrogenation and deoxygenation of bio-derived hydrocarbons, fatty acid esters, and free fatty acids; the only approved bio-source is the *Botryococcus braunii* species of algae. HC-HEFA shall be comprised of hydroprocessed synthesized paraffinic kerosene. The HC-HEFA synthetic blending components shall conform to the requirements in ASTM D7566 Annex A7. The finished fuel shall conform to the properties listed in tables I, II, IV and V and contain additives in accordance with 3.3 through 3.3.6.

3.2 Finished fuel. Finished fuels shall meet the requirements of this specification.

3.3 Additives. The finished fuel shall contain the type and amount of each additive specified in the acquisition document (see 6.2); the type and amount of each additive used shall be reported.

3.3.1 Antioxidants. Immediately after processing (i.e., during the rundown into feed/batch tank) and before the fuel is exposed to the atmosphere, an approved antioxidant shall be added to prevent the formation of gums and peroxides after manufacture. The concentration of antioxidant to be added shall be between 17.2 mg and 24.0 mg of active ingredient per liter of fuel (6.0 to 8.4 lbs/1000 barrels).

3.3.1.1 Formulations. The following antioxidant formulations are approved:

a. 2,6-di-tert-butyl-4-methylphenol
b. 6-tert-butyl-2,4-dimethylphenol
c. 2,6-di-tert-butylphenol
d. 75 percent min 2,6-di-tert-butylphenol
   25 percent max tert-butylphenols and tri-tert-butylphenols
e. 72 percent min 6-tert-butyl-2,4-dimethyhenol
   28 percent max tert-butyl-methylphenols and tert-butyl-dimethylphenols
f. 55 percent min 2,4-dimethyl-6-tert-butylphenol and 15 percent min 2,6-di-tert-butyl-4-methylphenol and 30 percent max mixed methyl and dimethyl tert-butylphenols

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

3.3.2 Metal deactivator. Metal deactivator additive shall not be used unless specified in the acquisition documents (see 6.2). A metal deactivator may be used if approved by the procuring activity and the user. Written consent for the use of metal deactivator shall be obtained from the NAVAIR Fuels and Lubricants Technical Warrant Holder (see 6.7). If approved, the metal deactivator, N,N'-disalicylidene-1,2-propanediamine, shall be blended into the fuel. The concentration of active material used on initial batching of the fuel at the refinery shall not exceed 2.0 mg/L. Cumulative addition of metal deactivator when redoping the fuel shall not exceed 5.7 mg/L. The final concentration of metal deactivator in the fuel shall be reported.

3.3.3 Corrosion inhibitor/lubricity improver. A corrosion inhibitor/lubricity improver additive in accordance with MIL-PRF-25017 shall be blended into JP-5. The amount added shall be equal to or greater than the minimum effective concentration and shall not exceed the maximum allowable concentration for an approved source as specified in QPL-25017. The point of injection of the corrosion inhibitor/lubricity improver shall be as specified in the acquisition documents (see 6.2).

3.3.4 Fuel system icing inhibitor. A fuel system icing inhibitor in accordance with MIL-DTL-85470 shall be added to JP-5 at the concentrations specified in table I. The point of injection of the additive shall be as specified in the acquisition documents (see 6.2).

3.3.5 Static dissipater additive. Static dissipater additive (SDA) shall not be used in JP-5 unless written consent has been obtained from the NAVAIR Fuels and Lubricants Technical Warrant Holder (see 6.7). If approved, the NAVAIR Fuels and Lubricants Technical Warrant Holder shall provide the allowable additive brand, additive dosage rate, and finished fuel electrical conductivity in accordance with ASTM D2624 requirements; the additive brand, concentration, and electrical conductivity shall be reported. Since table II does not define a minimum micro-separometer rating for JP-5 containing SDA, a micro-separometer limit shall also be defined at the time of approval. The concentration of electrical conductivity improver additive can be measured in accordance with ASTM D7524.

3.3.6 Premixing of additives. Additives shall not be premixed with other additives before injection into the fuel to prevent possible reactions among the concentrated forms of different additives.

3.4 Minimum finished fuel chemical and physical property requirements. The chemical and physical properties of all finished fuels shall meet the requirements specified in table I when tested in accordance with the specified test methods. Refer to 3.5 for additional chemical and physical property requirements for fuels containing co-hydroprocessed synthesized hydrocarbons (see 3.1.1) and synthetic blending components (see 3.1.2 to 3.1.7). The minimum micro-separometer rating of all finished fuels is dependent on the additives present (see table II). At the time of Government acceptance, the concentration of incidental contaminants in all finished fuels shall be within table V limits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

TABLE I. Chemical and physical property requirements and test methods.

| Property | Min | Max | Test Methods[1/] Referee | Alternative |
|---|---|---|---|---|
| **Composition** | | | | |
| Color, Saybolt | Report | | D156 | D6045 |
| Total acid number, mg KOH/g | | 0.015 | D3242 | |
| Aromatics, vol. % | | 25.0 | D1319 | D8305[2/], D8267 |
| | | 26.5 | | D6379[3/] |
| Sulfur, total, mass % | | 0.20 | D4294 | D2622, D5453 |
| Sulfur, Mercaptan, mass % or | | 0.002 | D3227 | |
| Doctor Test[4/] | | Negative | D4952 | |
| **Volatility** | | | | |
| Distillation temperature, °C | | | D86[5/] | D2887[6/] |
| Initial boiling point | Report | | | |
| 10 % recovered | | 205 | | |
| 20 % recovered | Report | | | |
| 50 % recovered | Report | | | |
| 90 % recovered | Report | | | |
| Final boiling point | | 300 | | |
| Residue, vol. % (D86 only) | | 1.5 | | |
| Loss, vol. % (D86 only) | | 1.5 | | |
| Flash point, °C | 60.0 | | D93 | D56[7/] |
| Density | | | D4052 | D1298, D7777 |
| Density, kg/L at 15°C | 0.788 | 0.845 | | |
| Gravity, API at 60°F | 48.0 | 36.0 | | |
| **Fluidity** | | | | |
| Freezing point, °C | | -46 | D2386 | D5972, D7153, D7154 |
| Viscosity at -20 °C, $mm^2/s$ | | 7.0 | D445 | D7042[8/], D7945 |
| **Combustion** | | | | |
| Net heat of combustion, MJ/kg | 42.6 | | D4809 | D3338/D3338M[9/], D4529[9/] |
| Hydrogen content, mass % | 13.4 | | D7171 | D3701, D5291 |
| Smoke point, mm or | 25.0 | | D1322 | |
| Smoke point, mm | 18.0 | | D1322 | |
| and naphthalene, vol % | | 3.0 | D1840 | D8305[10/] |
| Calculated Cetane Index[11/] | Report | | D976 | D4737 |
| **Corrosion** | | | | |
| Copper strip corrosion, 2 hr at 100°C | | No.1 | D130 | |

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

TABLE I. Chemical and physical property requirements and test methods – Continued.

| Property | Min | Max | Test Methods[1/] Referee | Alternative |
|---|---|---|---|---|
| **Thermal Stability** | | | | |
| Thermal Stability (2.5 hr at 260 °C) | | | | |
| Change in pressure drop, mmHg | | 25 | D3241 | |
| Tube deposit color code | | <3[12/] | | D3241 Annex A1 |
| or | | | | |
| Average deposit thickness rating, nm average over area of 2.5 $mm^2$ | | 85 | D3241 Annex A3[13/] | D3241 Annex A2 |
| **Additives** | | | | |
| Fuel system icing inhibitor, vol. % | 0.08 | 0.11 | D5006[14/] | |
| Corrosion Inhibitor/Lubricity Improver, mg/L | Report[15/] | | | |
| Antioxidants, mg/L | Report[16/] | | | |
| **Contaminants** | | | | |
| Existent gum, mg/100 mL | | 7 | D381 | IP 540[17/] |
| Micro-Separometer Rating | [18/] | | D3948 | |

1/ Where ASTM and IP methods are jointed, corresponding IP methods are accepted.

2/ Results from ASTM D8305 shall be bias-corrected using the bias-correction equation for total aromatics in Section 13 (Precision and Bias) of Test Method D8305. The bias-corrected aromatics result shall also be used in Test Method D3338.

3/ When using ASTM D6379 results, the higher maximum total aromatics limit shall apply.

4/ If the Doctor Test results in a failure ('positive' result), then mercaptan sulfur content shall be determined by the referee test method ASTM D3227.

5/ A condenser temperature of 0 °C to 5 °C shall be used for the distillation of JP-5 fuel.

6/ Distillation property criteria are specified in ASTM D86 scale units. ASTM D2887 results shall be converted to estimated ASTM D86 results by application of the correlation in Appendix X4 of ASTM D2887 for comparison with the specified property criteria. Distillation residue and loss limits provide control of the distillation process during the ASTM D86 test method and do not apply to ASTM D2887.

7/ ASTM D56 may give results up to 1 °C below the ASTM D93 results.

8/ Test Method D7042 results shall be converted to bias-corrected kinematic viscosity results by the application of the correction described in Test Method D7042 for jet fuel at –20 °C.

9/ For all grades use either Eq 1 or Table 1 in Test Method D4529 or Eq 2 in Test Method D3338. Calculate and report the net heat of combustion corrected for the sulfur content when using Test Method D4529 and D3338 empirical test methods.

10/ Results from D8305 shall be bias-corrected using the bias-correction equation for total polynuclear aromatics in Section 13 (Precision and Bias) of Test Method D8305.

11/ Boiling recovery temperatures may be obtained by either ASTM D86 or ASTM D2887 to perform the Cetane Index calculation. If ASTM D86 values are used, they shall be corrected to standard barometric pressure.

12/ If the visual rating of the heater tube shows peacock (P) or abnormal (A) type deposits, the fuel fails the tube deposit color code requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

13/ In cases of dispute between visual and metrological tube rating methods, the referee shall be Annex A3 ETR if available, otherwise Annex A2 ITR.
14/ See Note 2 on analog HB refractometer in ASTM D5006.
15/ See 3.3.3 for approved corrosion inhibitor/lubricity improver additives and dosage rates.
16/ See 3.3.1 for approved antioxidant additives and dosage rates.
17/ The preferred vaporizing medium for aviation turbine fuel is steam, however, the existent gum test IP 540 may be performed using air as the vaporizing medium. If air is used instead of steam, it shall be recorded. In case of a failure with air, the sample shall be retested using steam. Test Method ASTM D381 using steam jet operating conditions shall be the referee test method.
18/ The minimum micro-separometer rating using a micro-separometer (MSEP) instrument shall be as specified in table II. If the use of SDA is approved (see 3.3.5), the minimum micro-separometer rating shall be defined at the time of approval.

TABLE II. Micro-separometer rating.

| Additives* | MSEP Rating, min |
|---|---|
| Antioxidant (AO)*, Metal Deactivator (MDA)* | 90 |
| AO*, MDA*, and Fuel System Icing Inhibitor (FSII) | 85 |
| AO*, MDA*, and Corrosion Inhibitor/Lubricity Improver (CI/LI) | 80 |
| AO*, MDA*, CI/LI, and FSII | 70 |

*Even though the presence or absence of these additives does not change these limits, samples submitted for specification conformance testing shall contain the same additives present in the refinery batch. Regardless of which minimum the refiner is required to meet, the refiner shall report the MSEP rating on a laboratory hand blend of the fuel with all additives required by the specification.

3.5 Additional requirements for finished fuels containing synthesized hydrocarbons. In addition to meeting table I and II requirements, finished fuels containing co-hydroprocessed synthesized hydrocarbons (see 3.1.1) shall meet the additional requirements specified in table III and finished fuels containing synthetic blending components (see 3.1.2 to 3.1.7) shall meet the additional requirements specified in table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

TABLE III. Additional requirements for JP-5 containing co-hydroprocessed synthesized hydrocarbons.

| Property | Min | Max | Test Methods[1/] Referee | Test Methods[1/] Alternative |
|---|---|---|---|---|
| **Fluidity** | | | | |
| Viscosity at -40 °C, $mm^2/s$ | | 12.0 | D445 | D7945 |
| Freezing point, °C | | -46 | D5972[2/] | D7153, D7154 |
| **Thermal Stability** | | | | |
| Thermal Stability (2.5 hr at 280 °C) | | | | |
| Change in pressure drop, mmHg | | 25 | D3241 | |
| Tube deposit color code | | <3[3/] | | D3241 Annex A1 |
| or | | | | |
| Average deposit thickness rating, nm average over area of 2.5 $mm^2$ | | 85 | D3241 Annex A3[4/] | D3241 Annex A2 |
| **Contaminants** | | | | |
| Unconverted esters and fatty acids mg/kg[5/] | | 15 | D7797/IP 583[6/] | |

1/ Where ASTM and IP methods are jointed, IP methods are acceptable.

2/ Referee and alternative test methods vary from Table I when JP-5 contains co-hydroprocessed synthesized hydrocarbons. Inaccurate results may be obtained when using ASTM D2386 with fuels containing co-hydroprocessed synthesized hydrocarbons.

3/ If the visual rating of the heater tube shows peacock (P) or abnormal (A) type deposits, the fuel fails the tube deposit color code requirement.

4/ In cases of dispute between visual and metrological tube rating methods, the referee shall be Annex A3 ETR if available, otherwise Annex A2 ITR.

5/ Requirement is only applicable to fuels containing co-hydroprocessed synthesized kerosene derived from co-processing of esters and fatty acids as defined in section 3.1.1.1.

6/ In addition to fatty acid methyl esters (FAME), ASTM D7797 and IP 585 are able to identify carbonyl containing compounds such as those present in unreacted ester and fatty acid feedstocks. The reported value may be corrected for trace carbonyl species inherent in aviation turbine fuel derived from conventional sources. Corrected values shall be noted as such.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

TABLE IV. Additional requirements for JP-5 containing synthetic blending components.

| Property | Min | Max | Test Methods[1/] Referee | Test Methods[1/] Alternative |
|---|---|---|---|---|
| **Composition** | | | | |
| Aromatics, vol. % | 8.0<br>8.4 | | D1319 | D8305[2/], D8267<br>D6379[3/] |
| **Volatility** | | | | |
| Distillation temperature, °C | | | D86 | D2887[4/] |
| T50-T10 | 15 | | | |
| T90-T10 | 40 | | | |
| **Fluidity** | | | | |
| Viscosity at -40 °C, $mm^2/s$ | | 12.0 | D445 | D7945 |
| **Combustion** | | | | |
| Derived Cetane Number | 40 | | D6890 | D7668 |

1/ Where ASTM and IP methods are jointed, IP methods are acceptable.
2/ Results from ASTM D8305 shall be bias-corrected using the bias-correction equation for total aromatics in Section 13 (Precision and Bias) of Test Method D8305. The bias-corrected aromatics result shall also be used in Test Method D3338.
3/ When using ASTM D6379 results, the higher minimum total aromatics limit shall apply.
4/ ASTM D2887 results shall be converted to estimated ASTM D86 results by application of the correlation in Appendix X4 of ASTM D2887 for comparison with the specified property criteria.

3.6 Workmanship. At the time of Government acceptance, the finished fuel shall be clear and bright and visually free from undissolved water, sediment, or suspended matter as determined in accordance with ASTM D6986. In case of dispute, the fuel shall be clear and bright at 21 °C and shall contain no more than 1.0 mg/L of particulate matter in accordance with ASTM D2276 or ASTM D5452.

3.7 Incidental contaminants. At the time of Government acceptance, the concentration of incidental contaminants in the finished fuel shall not exceed the limits in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

TABLE V. Incidental contaminants.

| Property | Min | Max | Test Methods[1/] Referee | Alternative |
|---|---|---|---|---|
| Filtration Time, min[2/] | | 15 | MIL-DTL-5624 Appendix A | |
| Particulate matter (Gravimetric), mg/L[2/] or | | 1.0 | D5452 | D2276 |
| Particle Counting[3/] | | | D7619 | IP 565, IP 577 |
| ISO 4406 Code[4/] | | | | |
| ≥ 4 µm (c) | | 19 | | |
| ≥ 6 µm (c) | | 17 | | |
| ≥ 14 µm (c) | | 14 | | |
| ≥ 21 µm (c) | Report | | | |
| ≥ 25 µm (c) | Report | | | |
| ≥ 30 µm (c) | | 13 | | |
| Fatty Acid Methyl Ester (FAME),[5/] mg/kg | | 50 | IP 585 | D7797, IP 583, IP 590, IP 599 |

1/ Where ASTM and IP methods are jointed, IP methods are accepted.

2/ A minimum sample size of 3.785 liters (1 gallon) shall be filtered. Filtration time shall be determined in accordance with the procedure in Appendix A of this specification.

3/ Particle counting may be used as an alternate to the gravimetric particulate matter requirement. If any one of the ISO 4406 code limits is exceeded, particle counting fails and gravimetric particulate matter shall be the referee.

4/ The notation (c), used with particle sizes, denotes that the apparatus has been calibrated in accordance with ISO 11171.

5/ For a definition of FAME, see 6.6.5. The recent mandatory and voluntary introduction of FAME in the commercial middle distillate marketplace has resulted in the potential for trace amounts of FAME in JP-5 fuel. Fuel supplied under this specification shall not intentionally be blended with FAME. Where a risk exists for incidental FAME contamination, the supplier shall ensure this limit is not exceeded prior to product availability. Fatty acid methyl esters that fail to meet the biodiesel quality standards are not permitted in aviation turbine fuel.

3.8 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

## 4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall consist of all examinations, inspections, and tests of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

4.1.1 Inspection lot. For conformance inspection, individual lots shall be examined, inspected, and tested as specified herein to ensure individual lots meet all the requirements specified in section 3.

4.1.2 Sampling plans.

4.1.2.1 Sampling for conformance inspection. Each bulk or packaged lot (see 6.6) of material shall be sampled in accordance with ASTM D4057 or ASTM D4177 except where individual test procedures contain specific sampling instructions.

4.1.2.1.1 Sample containers. Examine the sample container for conformance to recommended sample container guidance found in ASTM D4306 (see 6.5).

4.1.2.2 Sampling for examination of filled containers for delivery. A random sample of filled containers shall be selected from each lot. The samples shall be examined in accordance with 4.3.1.3.

4.2 Inspection conditions. The finished fuel shall meet the limiting values in table I using the specified test methods. In addition to meeting table I requirements, finished fuels containing co-hydroprocessed synthesized hydrocarbons (see 3.1.1) shall meet the additional requirements specified in table III and finished fuels containing synthetic blending components (see 3.1.2 to 3.1.7) shall meet the additional requirements specified in table IV.

4.3 Methods of inspection.

4.3.1 Examination of product.

4.3.1.1 Visual inspection. Samples selected in accordance with 4.1.1 shall be visually examined for compliance with 3.6.

4.3.1.2 Examination of empty containers. Prior to filling, each empty unit container shall be visually inspected for cleanliness and prepared for proper usage in accordance with ASTM D4057.

4.3.1.3 Examination of filled containers. Samples taken as specified in 4.1.2 shall be examined for conformance to MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing, and markings.

4.3.2 Chemical and physical tests. Tests to determine conformance to chemical and physical requirements shall be conducted in accordance with table I, II, and V for all finished fuels. Finished fuels containing co-hydroprocessed synthesized hydrocarbons (see 3.1.1) shall also meet the additional requirements specified in table III and finished fuels containing synthetic blending components (see 3.1.2 to 3.1.7) shall also meet the additional requirements specified in table IV. Requirements contained in tables I, II, III, IV, and V are not subject to corrections for test tolerances. If multiple determinations are made, results falling within any specified repeatability and reproducibility tolerances shall be averaged to determine conformance to the specified limit. The following applies to all specified limits in this standard

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

with exception of flash point: For purposes of determining conformance with this specification, an observed value or a calculated value shall be rounded "to the nearest unit" in the last right-hand digit used in expressing the specification limit, in accordance with the rounding method of ASTM E29. The flash point value is an absolute limit as defined in ASTM E29.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the acquisition documents (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. JP-5 covered by this specification is intended for use in aircraft turbine engines. JP-5 requires military-unique additives that are necessary in military weapon systems, which is unique to military aircraft, engine designs, and missions. Additionally, JP-5 is a military-unique fuel because it is required to have a substantially higher flash point than commercial aviation turbine fuels for shipboard safety. Since, JP-5 is stored in large quantities on aircraft carriers and other vessels, the higher flash point is necessary for safety in these military-unique applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Grade of fuel (JP-5).
c. Identification of type and concentration of synthetic material present in the fuel. As applicable, documentation certifying compliance of synthetic blending components with the appropriate ASTM D7566 annex requirements (see 3.1).
d. Information concerning the type and amount of each additive used (see 3.3).
e. Location and injection method of the corrosion inhibitor/lubricity improver (see 3.3.3).
f. Location and injection method of the fuel system icing inhibitor (see 3.3.4).
g. If approved, location and injection method of the metal deactivator additive (see 3.3.2).
h. If approved, location and injection method of the static dissipater additive (see 3.3.5).
i. Quantity required and size containers desired.
j. Level of packaging and packing required (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

6.3 Conversion of metric units. Units of measure have been converted to the International System of Units (SI) (metric) in accordance with IEEE/ASTM SI 10 “American National Standard for Metric Practice.” If test results are obtained in units other than metric or there is a requirement to report dual units, IEEE/ASTM SI 10 should be used to convert the units.

6.4 Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313. During transition to Globally Harmonized System (GHS) safety data sheets, refer to OSHA guidance.

6.5 Sample containers. A number of jet fuel properties are very sensitive to trace contamination from sample containers. Refer to ASTM D4306 for recommended sample containers.

6.6 Definitions. The definitions listed in this section are specific to this document. For a definition of additional terminology relating to petroleum products and liquid fuels refer to ASTM D4175.

6.6.1 Bulk lot. A bulk lot consists of an indefinite quantity of a homogeneous mixture of material offered for acceptance in a single isolated container or manufactured in a single plant run through the same processing equipment, with no change in ingredient material.

6.6.2 Packaged lot. A packaged lot consists of an indefinite number of 208-liter (55-gallon) drums or smaller unit packages of identical size and type, offered for acceptance, and filled from the isolated tank containing a homogeneous mixture of material, or filled with a homogeneous mixture of material run through the same processing equipment, with no change in ingredient material.

6.6.3 Homogeneous product. A homogeneous product is defined as a product where samples taken at various levels of the batch tank are tested for the defining homogeneous characteristics and all values obtained meet the repeatability precision requirements for that test method.

6.6.4 Finished fuel. Final blend of a complex mixture of hydrocarbons, with additives, provided for specification acceptance.

6.6.5 Fatty acid methyl ester (FAME). FAME is synonymous with biodiesel meeting the requirements of ASTM D6751 or EN 14214. As specified in ASTM D6751, "biodiesel is a fuel comprised of mono-alkyl esters of long chain fatty acids derived from vegetable oils or animal fats, designated B100."

6.6.6 Synthesized paraffinic kerosene (SPK). Kerosene consisting of n-paraffins, iso-paraffins, and cycloparaffins derived from non-petroleum sources.

6.6.7 Conventional blending component. Blending streams derived from the following

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

conventional sources: crude oil, natural gas liquid condensates, heavy oil, petroleum, oil sands, oil shale, or mixtures thereof.

6.6.8 Synthetic blending component. Blending streams derived from non-conventional sources meeting the requirements of 3.1.2, 3.1.3, 3.1.4, 3.1.5, 3.1.6, and 3.1.7. Synthetic blending components must be combined with a conventional blending component prior to certification.

6.6.9 Co-hydroprocessed synthesized kerosene. Hydrocarbons in the kerosene boiling range derived from non-petroleum sources such as coal, natural gas, biomass, fatty acid esters, and fatty acids by processes such as gasification, FT synthesis, and hydroprocessing that have been processed simultaneously with hydrocarbons from conventional sources.

6.6.10 Hydroprocessed esters and fatty acids (HEFA) SPKs. A synthetic blending component comprised of SPK produced by hydroprocessing plant and algal oils, or animal fats.

6.6.11 Fischer-Tropsch hydroprocessed synthesized paraffinic kerosene (FT-SPK). A synthetic blending component comprised of SPK produced from one or more precursors synthesized by Fischer-Tropsch processing.

6.6.12 Synthesized iso-paraffins (SIP). A synthetic blending component comprised of primarily iso-paraffins (farnesane) that is produced by hydroprocessing and fractionation of farnesene derived from fermentation of sugars.

6.6.13 Fischer-Tropsch synthesized paraffinic kerosene plus aromatics (SPK/A). A synthetic blending component comprised of SPK and synthesized aromatic compounds.

6.6.14 Alcohol-to-jet synthetic paraffinic kerosene (ATJ-SPK). A synthetic blending component comprised of SPK from alcohol via dehydration, oligomerization, hydrogenation, and fractionation processing steps.

6.6.15 Synthesized paraffinic kerosene from hydroprocessed hydrocarbons, esters, and fatty acids (HC-HEFA). A synthetic blending component that is comprised of bio-derived hydrocarbons and free fatty acids and fatty acid esters that have been hydroprocessed to saturate the hydrocarbon molecules and remove essentially all oxygen.

6.7 NAVAIR approval. To obtain written consent, contact the NAVAIR Fuels and Lubricants Technical Warrant Holder, BLDG 2360, Propulsion Systems Evaluation Facility, 22229 Elmer Road, Patuxent River, MD 20670-1534.

6.8 Subject term (key word) listing.

Alcohol-to-jet synthetic paraffinic kerosene (ATJ-SPK)
Antioxidants
Co-hydroprocesed synthesized paraffinic kerosene
Corrosion inhibitor
Fischer-Tropsch hydroprocessed synthesized paraffinic kerosene (FT-SPK)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

Fischer-Tropsch synthesized paraffinic kerosene plus aromatics (SPK/A)
Flash point
Freezing point
Fuel system icing inhibitor
Hydroprocessed esters and fatty acids (HEFA)
Hydrocarbon distillate
Hydrogen content
Lubricity improver
Static dissipater additive
Synthesized iso-paraffins (SIP)
Synthesized paraffinic kerosene (SPK)
Synthesized paraffinic kerosene from hydroprocessed hydrocarbons, esters, and fatty acids (HC-HEFA)

6.9 International standardization agreement implementation. This specification implements NATO STANAG 1135 "Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations," NATO AFLP 3747 "Guide Specifications (Minimum Quality Standards) for Aviation Turbine Fuels (F-24, F-34, F-35, F-37, F-40 and F-44)," and AFIC AIR STD FG 4024 “Interchangeability Chart of Standardised Aviation Fuels, Lubricants, and Associated Products.” When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

# METHOD FOR DETERMINATION OF FILTRATION TIME AND TOTAL SOLIDS (PARTICULATE)

## A.1 SCOPE

A.1.1 Scope. This method describes a procedure to determine singularly or simultaneously the filterability characteristics and solids contamination of jet fuel. The purpose is to detect and prevent contaminants in jet fuel, which can plug and cause rupture of ground filtration equipment, thereby affecting flight reliability/safety of aircraft. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

## A.2 METHODS

A.2.1 Summary of method. An amount of 3.785 liters (1 gallon) of jet fuel is filtered through a membrane filter in the laboratory. The time required to filter this volume is measured in minutes and solids content is determined gravimetrically.

## A.3 APPARATUS

a. Membrane filter: White, plain 47 mm diameter, nominal pore size 0.8 micron. The membrane filter shall conform to the ASTM D5452 requirements.

b. Filtration apparatus: The apparatus, constructed of stainless steel, consists of a funnel and funnel base with a filter support such that a membrane filter can be securely held between the sealing surface of the funnel and the funnel base (see ASTM D5452, Figure 1).

c. Vacuum flask: A minimum of 4 liters.

d. Vacuum system: That develops in excess of 67.5 kPa (20 in. of mercury) vacuum.

e. Oven: Of the static type (without fan assisted circulation) controlling to 90 °C ±5 °C.

f. Forceps: Flat-bladed with non-serrated, non-pointed tips.

g. Dispenser, rinsing solvent (petroleum ether): Containing a 0.45 micron membrane filter in the delivery line. If the solvent has been pre-filtered using a 0.45 micron filter, then an inline filter is not required.

h. Glass Petri dish: Approximately 125 mm in diameter with removable cover.

i. Analytical balance: Single or double pan, the precision standard deviation of which shall be 0.07 mg or better.

## A.4 PREPARATION

A.4.1 Preparation of apparatus and sample containers. All components of the filtration apparatus (except the vacuum flask), sample containers, and their caps shall be cleaned as described in ASTM D5452. All metal parts of the filtration apparatus are to be electrically bonded and grounded, including the fuel sample container. See ASTM D5452 for other safety precautions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

## A.5 SAMPLING

A.5.1 Sample. Obtain a representative 3.785 liters (1 gallon) sample as directed in ASTM D5452. When sampling from a flowing stream is not possible, an all-level sample or an average sample in accordance with ASTM D4057 and/or ASTM D4177 shall be permitted. The 3.785 liter sample container shall be an interior epoxy-coated metal can, a brown glass bottle, or a clear glass bottle protected by suitable means from exposure to light.

## A.6 PROCEDURE

A.6.1 Test procedure.

a. Using forceps, place a new membrane (test) filter in a clean petri dish. Place the petri dish with the lid slightly ajar in a 90 °C ± 5 °C oven for 30 minutes. Remove the petri dish from the oven and place it near the balance with the lid slightly ajar, but still protecting the filter from airborne contamination, for 30 minutes.

b. Weigh the test filter. A filter weighing in excess of 90 mg shall not be used in the test.

c. Immediately prior to filtering the fuel, shake the sample to obtain a homogenous mix and ensure that fuel temperature does not exceed 30 °C. Clean the exterior or top portion of the sample container to ensure no contaminants are introduced. Any free water present in the fuel sample will invalidate the filtration time results by giving an excessive filtration time rating.

d. With the vacuum off, pour approximately 200 mL of fuel into the funnel.

e. Turn vacuum on and record starting time. Continue filtration of the 3.785-liter sample, periodically shaking the sample container to maintain a homogenous mix. Record the vacuum (kPa or inches of mercury) 1 minute after start and again immediately prior to completion of filtration. Throughout filtration, maintain a sufficient quantity of fuel in the funnel so the membrane filter is always covered.

f. Record the filtration time in minutes expressed to the nearest whole number. If filtration of the 3.785 liters is not completed within 30 minutes, the test shall be stopped, and the volume of the fuel filtered shall be measured. In these cases, report the filtration time as “greater than 30 minutes” and the total volume of fuel filtered.

g. Record the vacuum in kPa (in. of mercury) as determined from the average of the two readings taken in A.6.1.f.

h. After recording the filtration time, shut off the vacuum and rinse the sample container with approximately 100 mL of filtered petroleum ether and dispense into the filtration funnel. Turn on the vacuum and filter the 100 mL rinse. Turn off the vacuum and wash the inside of the funnel with approximately 50 mL of filtered petroleum ether. Turn on the vacuum and filter. Repeat the funnel rinse with another 50 mL of petroleum ether but allow the rinse to soak the filter for approximately 30 seconds before turning on the vacuum to filter the rinse. With the vacuum on, carefully remove the top funnel and rinse the periphery of the membrane filter by directing a gentle stream of petroleum ether from the solvent dispenser from the edge of the membrane toward the center, taking care not to wash contaminants off the filter. Maintain vacuum after final rinse for a few seconds to remove the excess petroleum ether from the filter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

i. Using forceps, carefully remove the test filter from the funnel base and place in a clean Petri dish. Dry in the oven at 90 °C ±5 °C for 30 minutes with the cover on the Petri dish slightly ajar. Remove the petri dish from the oven and place it near the balance with the lid slightly ajar, but still protecting the filter from airborne contamination, for 30 minutes. If more than one sample is processed, cooling time may have to be increased. Re-weigh the filter.

j. Record the total solids content in mg/liter by using the following formula:

$$\frac{\text{Weight gain of filter in mg}}{\text{3.785 L}} = \text{mg/L}$$

k. Should the sample exceed the 30-minute filtration time and a portion of the fuel is not filtered, the solids content in mg/liter shall be reported as follows: Determine the volume of fuel filtered by subtracting the mL of fuel remaining from 3785 mL.

$$\frac{\text{Weight gain of filter in mg}}{\text{mL of fuel filtered X 0.001}} = \text{mg/L}$$

A.7 LIMITS

A.7.1 Filtration time:

(1) The maximum allowable filtration time shall be 15 minutes.

(2) The vacuum shall exceed 67.5 kPa (20 inches of mercury) throughout the test, i.e., the differential pressure across the filter should exceed 67.5 kPa (20 inches of mercury).

(3) The fuel temperature shall be between 18 °C and 30 °C.

A.7.2 Total solids: Maximum allowable particulate matter is 1.0 mg/liter.

A.8 NOTES

A.8.1 If it is desired to determine the filtration time and not the total solids content, perform the test by omitting weighing steps, A.6.1j, and A.6.1k calculation.

A.8.2 If it is desired to determine the total solids content and not the filtration time, total solids shall be determined in accordance with ASTM D5452, and the use of a control filter shall be required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.

## CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - AS
Air Force – 68
DLA - PS

Preparing activity:
Navy - AS

(Project 9130-2024-001)

Review activities:
Army - AR, AV
Navy - SH
Air Force - 11

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:23Z
Check the source to verify that this is the current version before use.